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ECA-3102

NANO ELECTRONICS

Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

Aim:

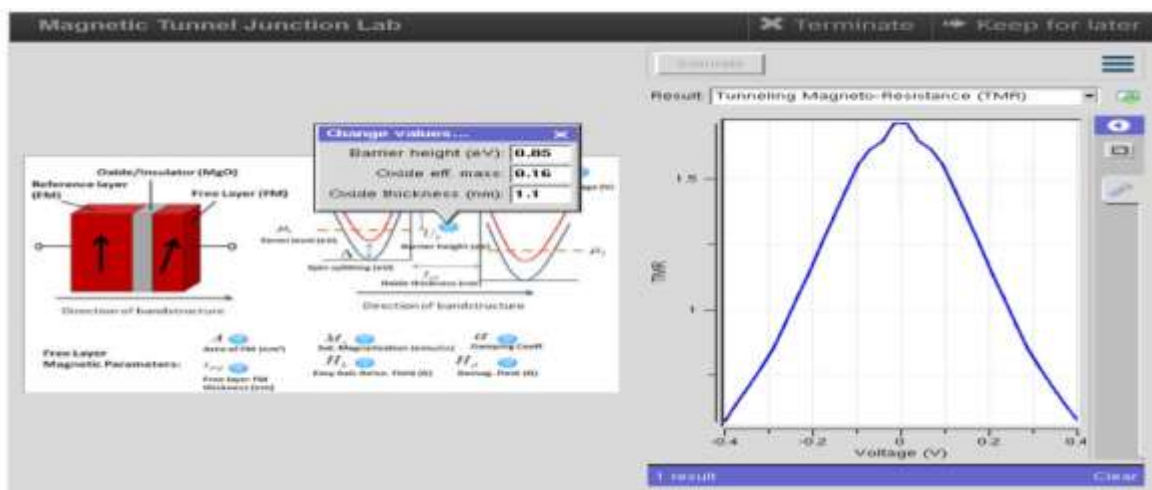
To obtain the Tunnelling Magnetoresistance (TMR) by varying barrier height. Software:

NanoHub Online Tube

Tools Required: Nanohub

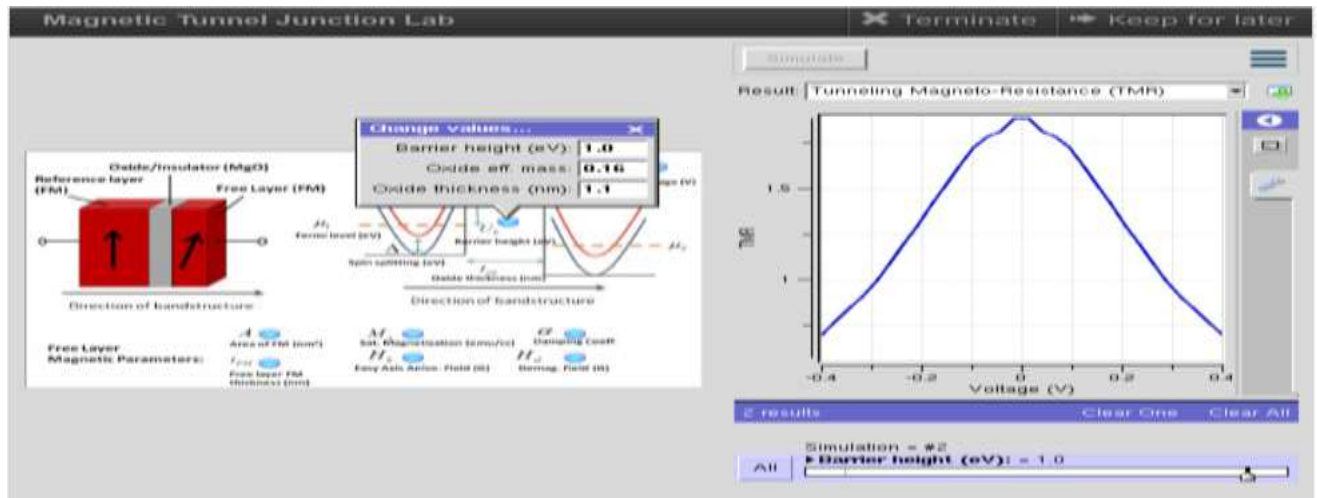
Output:

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



The image shows the **Magnetic Tunnel Junction (MTJ) Lab** simulation in **nanoHUB**. It displays the **Tunneling Magneto-Resistance (TMR)** graph for a **barrier height of 0.85 eV**. The graph shows that **TMR is highest at 0 V** and **decreases as the voltage increases in both positive and negative directions**, indicating reduced tunneling efficiency at higher voltages.

Case Study 2: Characteristics of TMR with barrier height of 1.0eV



This image shows the Magnetic Tunnel Junction (MTJ) Lab simulation with a barrier height of 1.0 eV. The graph displays Tunneling Magneto-Resistance (TMR) vs Voltage. The TMR reaches its maximum near 0 V and decreases as the voltage increases in both positive and negative directions, showing the voltage dependence of the MTJ.

Result: SIMULATION OF TUNNELLING MAGNETORESISTANCE (TMR) BY VARYING BARRIER HEIGHT IS ACCOMPLISHED.